

Combined Accentuated Eccentric Loading and Rest Redistribution in High-Volume Back Squat: Acute Kinetics and Kinematics

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Abstract

Chae, S, Long, SA, Lis, RP, McDowell, KW, Wagle, JP, Carroll, KM, Mizuguchi, S, and Stone, MH. Combined accentuated eccentric loading and rest redistribution in high-volume back squat: Acute kinetics and kinematics. *J Strength Cond Res* 38(4): 640–647, 2024—The purpose of this study was to explore acute kinetic and kinematic responses to combined accentuated eccentric loading and rest redistribution (AEL + RR). Resistance-trained men ($n = 12$, 25.6 ± 4.4 years, 1.77 ± 0.06 m, and 81.7 ± 11.4 kg) completed a back squat (BS) 1 repetition maximum (1RM) and weight releaser familiarization session. Three BS exercise conditions (sets $\times$ repetitions $\times$ eccentric/concentric loading) consisted of (a) $3 \times (5 \times 2) \times 110/60\%$ (AEL + RR 5), (b) $3 \times (2 \times 5) \times 110/60\%$ (AEL + RR 2), and (c) $3 \times 10 \times 60/60\%$ 1RM (traditional sets [TS]). Weight releasers (50% 1RM) were attached to every first repetition of each cluster set (every first, third, fifth, seventh, and ninth repetition in AEL + RR 5 and every first and sixth repetition in AEL + RR 2). The AEL + RR 5 resulted in significantly ($p < 0.05$) greater concentric peak velocity (PV) (1.18 ± 0.17 m·s⁻¹) and peak power (PP) ($2,304 \pm 499$ W) compared with AEL + RR 2 (1.11 ± 0.19 m·s⁻¹ and $2,148 \pm 512$ W) and TS (1.10 ± 0.14 m·s⁻¹ and $2,079 \pm 388$ W). Furthermore, AEL + RR 5 resulted in significantly greater PV and PP across all 10 repetitions compared with TS. Although AEL + RR 5 resulted in significantly greater concentric mean force (MF) ($1,706 \pm 224$ N) compared with AEL + RR 2 ($1,697 \pm 209$ N) and TS ($1,685 \pm 211$ N), no condition by set or repetition interactions existed. In conclusion, AEL + RR 5 increases PV and PP but has little effect on MF. Coaches might consider prescribing AEL + RR 5 to increase especially peak aspects of velocity and power outcomes.

Key Words: eccentric overload, weight releaser, set structure, intraset rest, strength, power

Introduction

As athletes become adapted to conventional training, they might need novel training stimuli to optimize strength and power development (9,31). Accentuated eccentric loading (AEL) using eccentric overloading in coupled eccentric and concentric actions can acutely potentiate concentric kinetics and kinematics (20,40) and develop long-term strength and power (5,11,24). In the applied setting, the use of weight releasers (WR) seems to be a practical way of prescribing AEL for barbell exercises such as back squat (BS) compared with other equipment (10,14,41). When multiple AEL repetitions are performed in a set, alternative set structures (17,19,32) should be used, so that a pair of WR can be reattached to a barbell during intraset rests (38,39). Rest redistribution (RR) can be a practical alternative set structure when using AEL because it does not extend total rest period, unlike basic cluster sets (12,33), while acutely maintaining concentric kinetics and kinematics (17,19,32).

Merrigan et al. (22) and Wagle et al. (38,39) prescribed AEL for the first repetition of a set during 3 sets of 5 repetitions of BS exercise with (eccentric/concentric) 120/65% and 105/80% BS 1 repetition maximum (1RM), respectively. In a work by Merrigan et al. (22),

AEL potentiated the concentric force, velocity, and power until the third–fourth repetition, whereas Wagle et al. (38,39) did not report any concentric potentiation. Researchers who determined acute mechanical responses to RR used 3–4 sets of 10–12 repetitions of BS exercise with 65–75% BS 1RM and 1–6 cluster repetitions (1,16,18,23,25–28,34,35). It has been observed that RR acutely maintained concentric velocity and power at later 1–5 cluster repetitions (1,16,23,25–28,34,35) but to a lesser magnitude with 6 cluster repetitions (18). The effect of both potentiation (22) and maintenance (1,16,18,23,25–28,34,35) of concentric force, velocity, and power seems to exist until the fourth or fifth repetition. Therefore, using AEL for the first repetition of less than or equal to 2–5 cluster repetitions using RR might potentiate and maintain concentric force, velocity, and power throughout most repetitions during BS exercise.

In a strength-endurance phase of block periodization, athletes usually perform high-volume exercises (e.g., 3 sets of 10 repetitions) to develop work capacity and hypertrophy (2,3). Traditional sets (TS) that use isoinertial eccentric/concentric loading without intraset rests cause a substantial decrease in concentric velocity and power during high-volume BS exercise (1,18,23). Kinetics and kinematics including concentric mean force (MF), mean velocity (MV), mean power (MP), peak force (PF), peak velocity (PV), peak power (PP), impulse (IMP), and rate of force development (RFD) have all been known to be potent training

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stimuli for strength and power adaptations (8,36,37). Given the considerable decrease in concentric velocity and power, TS might, in part, compromise potent training stimuli during high-volume BS exercise compared with AEL (22) or RR (1,16,23,25–28,34,35). However, combined accentuated eccentric loading and rest redistribution (AEL + RR) might obviate the downside associated with TS during high-volume BS exercise while not decreasing the potential for work capacity and hypertrophy in a strength-endurance phase (2,3).

As discussed in the second paragraph, potentiation and maintenance effect may occur when AEL is prescribed for the first repetition of 2–5 cluster repetitions. Particularly, during 3 sets of 10 repetitions of BS exercise, it may be systematic to prescribe AEL for 2 and 5 cluster repetitions to determine the effect of the number of AEL repetitions at each end. Furthermore, given that Merrigan et al. (22), who reported the potentiation effect with 120/65% BS 1RM, prescribed AEL for the initial repetition only during low-volume exercise, it might be wise to decrease the load to consider single vs. multiple AEL repetitions and low volume vs. high volume. Detailed rationales for the selection of certain eccentric/concentric loading for the current study are described in the “Methods” section.

The purpose of this study was to investigate acute mechanical responses between 2 AEL + RR conditions and TS during 3 sets of 10 repetitions of BS exercise in resistance-trained men. This study considered 3 practical factors: (a) BS exercise, (b) WR, and (c) RR. First, the BS exercise is a widely used closed kinetic chain and multijoint lower-body exercise that has a great potential to confer desired alterations (13). Second, WR involve relatively low budget and spatial requirements in prescribing AEL (20,40). Third, RR is a subclass of alternative set structures that could minimize concerns for time constraints considering a lack of extended total rest period (17,19,32). This study also measured a range of acute kinetics and kinematics, so that coaches can be informed as to which condition is potentially effective for force production, velocity aspect, and power output. Based on relevant studies, it was hypothesized that 2 AEL + RR conditions would

result in greater concentric force, velocity, and power compared with TS. In addition, it was hypothesized that AEL + RR that prescribed 5 AEL repetitions would cause the greatest concentric force, velocity, and power among all 3 conditions, given potentially more frequent potentiation and maintenance across 10 repetitions.

Methods

Experimental Approach to the Problem

This study used a within-subject, randomized, counterbalanced, and repeated-measures design. Subjects completed a session of concentric BS 1RM testing and WR familiarization on the first visit. Subjects completed 3 exercise conditions (Figure 1) on the second, third, and fourth visit in a randomized and counterbalanced manner. Each visits was separated by 7 days.

During the concentric phase, MF, MV, MP, IMP, PF, PV, PP, and RFD were measured for each repetition using linear position transducers with dual force plates (details are described in the “Data Collection” subsection). To warrant the ecological validity, subjects maintained their normal unsupervised diet and training that were not registered. Subjects, however, were asked to avoid (a) alcohol for 48 hours, (b) exercise for 24 hours, and (c) caffeine for 8 hours before each visit.

Subjects

Twelve resistance-trained adult men (a range of 19–34) participated in this study (Table 1). Inclusion criteria were (a) adult men without limitations in BS exercise, (b) a proficient BS exercise technique, and (c) ≥ 1 year of BS training experience. Two subjects indicated their limited previous experience performing AEL using WR. However, all subjects were familiarized with the execution of AEL using WR before the first testing session. Moreover, subjects only performed AEL exercise during supervised sessions (2 AEL + RR conditions). Before participating in this study, all subjects signed on written informed consent. This study

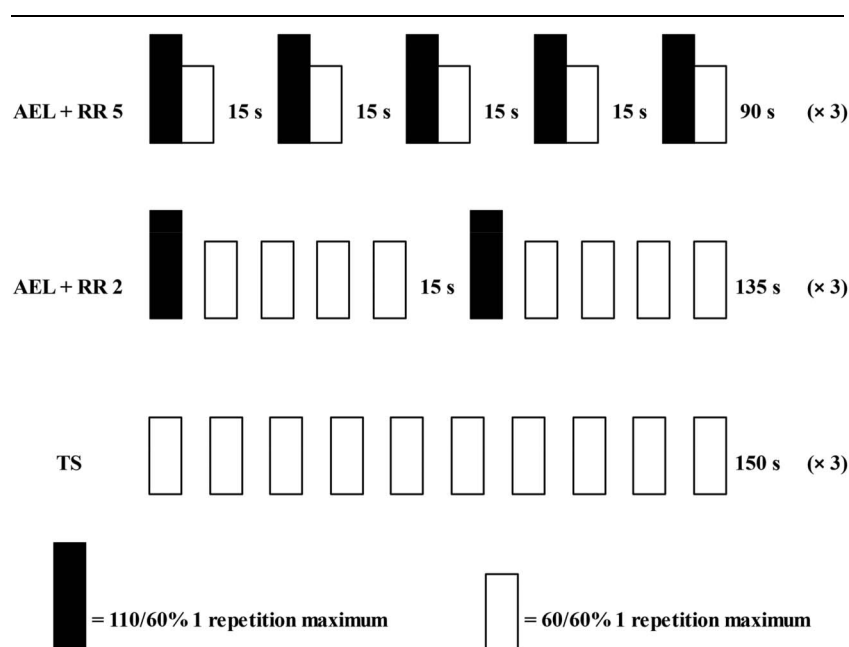


Figure 1. A schematic representation of loading schemes and set structures.

Table 1
Subjects' demographic information.*†

Age (y)	25.6 ± 4.4
Height (m)	1.77 ± 0.06
Body mass (kg)	81.7 ± 11.4
Fat-free mass (kg)	66.4 ± 6.9
Fat mass (kg)	15.4 ± 6.1
Fat mass (%)	18.4 ± 5.6
Skeletal muscle mass (kg)	33.3 ± 3.6
Resistance training experience (y)	8.1 ± 4.1
BS training experience (y)	7.8 ± 4.2
BS 1RM (kg)	152.4 ± 27.8
BS 1RM to body mass (kg·kg ⁻¹ body mass)	1.9 ± 0.4

*BS = back squat; 1RM = 1 repetition maximum.

†Data are presented as mean ± SD.

was approved by the East Tennessee State University's Institutional Review Board (approval number: 0322.2f).

Procedures

First Visit. On each visit before data collection, subjects' hydration status was measured (urine specific gravity was lower than 1.020) using a refractometer to ensure that hydration status would not affect the study results. Subjects' height and body mass were measured. Subjects' body composition was then determined using a bioelectrical impedance analyzer (BCA 514, Seca, Hamburg, Germany).

Subjects completed a standardized laboratory warm-up (25 repetitions of jumping jacks, 1 set of 5 repetitions of midhigh pull using a 20-kg bar, and 3 sets of 5 repetitions of midhigh pull using a 60-kg bar). Subsequently, subjects completed a session of BS 1RM measurement using the same warm-up procedures described by Wagle et al. (38,39) (Table 2). Subjects were instructed to lower the bar until the top of their thighs was parallel to the ground. This depth was kept consistent throughout the data collection. Subjects controlled their pace during eccentric phase and performed concentric phase as explosively as possible throughout the data collection. The loading scheme for 2 AEL + RR and TS conditions was determined based on subjects' concentric BS 1RM.

Ten to 15 minutes after BS 1RM measure, subjects completed a WR familiarization session using a pair of WR (Rogue Fitness, Columbus, OH) (Table 2). Briefly, appropriate WR height was determined for each subject where the WR were detached from the bar at the lowest BS depth with a minimal interference of natural transition from eccentric to concentric phases (40). Subjects were instructed to focus on the transition, stabilizing the bar, and walking out of the rack. The WR height and bar height were recorded on the first visit and kept consistent throughout the data collection.

Second, Third, and Fourth Visits. Seven days after the first visit, subjects completed a BS exercise condition at the same time with at most ±2 hours of window to reduce circadian influences, which otherwise might affect the study results. Subjects completed the same standardized laboratory warm-up and BS warm-up sets as used for the first visit. However, after the second warm-up set for the BS, subjects completed 2 different final warm-up sets with or without a pair of WR to equate total volume load (eccentric + concentric) across 3 exercise conditions (Table 2). The purpose of using the final warm-up set with a pair of WR in 2 AEL

+ RR conditions was to help subjects with movement practices using similar loading scheme to the experimental conditions. Then subjects completed experimental conditions.

Although not reported in this study, the authors concurrently measured acute metabolic and mechanical responses during exercise. The metabolic aspects of this investigation will be discussed in a follow-up article. After the final warm-up set, subjects put on breathing equipment (a nose clip, headgear, and mouthpiece, which was connected to breathing tube of metabolic cart). Subjects were wearing breathing equipment while performing exercise and taking rest.

During 2 AEL + RR conditions, subjects unracked the bar and WR that were attached to the bar (total 110% BS 1RM). When WR contacted the floor and were detached from the bar as subjects descended, they performed concentric phase with the bar only (60% BS 1RM). Subjects performed the following repetition(s) (1 and 4 more for AEL + RR 5 and AEL + RR 2, respectively) with the bar only in a consecutive manner, completing 5 and 2 AEL repetitions per set for AEL + RR 5 and AEL + RR 2, respectively. Once the bar was racked, subjects took (waited) 15 seconds of intraset and 90 or 135 seconds of interset rests corresponding to each condition. Investigators, 1 on each side, reapplied loaded WR (50% BS 1RM) to the bar during the intraset and interset rest. The 15 seconds of intraset rest period was long enough for the investigators to reattach the loaded WR to the bar. When 5 seconds remained, subjects were asked to position themselves under the bar, which was kept consistent across all conditions including TS. Once a rest period ended, subjects unracked the bar and walked out of the rack to complete the prescribed repetitions. All subjects were able to complete the prescribed loading, sets, and repetitions during all conditions.

To elicit a concentric potentiation effect while considering multiple AEL repetitions during high-volume exercise, we first chose 60% concentric loading because it roughly corresponds to

Table 2
Warm-up and WR familiarization procedures.*

BS 1RM session warm-up			
Set × repetition × load (% estimated BS 1RM)	Inter-set rest		
1 × 5 × 30/30% eccentric/concentric	60 s		
1 × 3 × 50/50% eccentric/concentric	60 s		
1 × 2 × 70/70% eccentric/concentric	120 s		
1 × 1 × 80/80% eccentric/concentric	180 s		
1 × 1 × 90/90% eccentric/concentric	180 s		
3–5 incremental BS 1RM trials	180 s		
WR familiarization			
Set × repetition	WR load	Bar load	Inter-set rest
1 × (2 × 5)†	Unload (about 16 kg)	Unload (20 kg)	60 s
1 × (5 × 2)†	Unload (about 16 kg)	Unload (20 kg)	60 s
5 × 1‡	50% BS 1RM	Unload (20 kg)	30 s
Experimental sessions warm-up			
Set × repetition × load (% BS 1RM)	Inter-set rest	Condition	
1 × 5 × 30/30% eccentric/concentric	60 s	All conditions	
1 × 3 × 50/50% eccentric/concentric	60 s	All conditions	
1 × 2 × 100/50% eccentric/concentric	180 s	AEL + RR conditions	
1 × 2 × 75/75% eccentric/concentric	180 s	TS condition	

*AEL + RR = combined accentuated eccentric loading and rest redistribution; BS 1RM = back squat 1 repetition maximum; TS = traditional sets; WR = weight releasers.

†WR were applied to the first repetition of cluster repetitions.

‡WR were applied to the first repetition of each set.

moderate relative intensity for 3 sets of 10 repetitions (2,3) and is within the range of optimal BS loading for maximal power (30). We then opted for 110% eccentric loading because 120% may be difficult for some subjects to control during decent when applied to multiple repetitions (21,22). Although 105% was tolerable when used for every repetition in a set (a total of 15), it did not seem to be enough for the potentiation effect (38,39). A gap between eccentric and concentric loading was 50%, which seems to be sufficient to result in the potentiation effect (21,22,38,39).

Data Collection. All mechanical data acquisition was completed using 4 linear position transducers (Celesco Measurement Specialties, Chatsworth, CA) synchronized with dual force plates ($2 \times 91 \times 45.5$ cm) (RoughDeck HP-H; Rice Lake Weighing Systems, Rice Lake, WI) through a custom LabView program (ver. 18; National Instruments Corp, Austin, TX) with the sampling frequency of 1,000 Hz. The 4 linear position transducers were attached to the top of a custom-built power rack, which is the same mechanical data acquisition method that enabled the calculation of vertical displacement described in previous work by Cormie et al. (6,7). To reduce noise from mechanical and electrical origins, a digital Butterworth fourth-order low-pass filter was used with the cutoff frequency of 58 Hz (42). Data were analyzed using a custom LabView program (ver. 18; National Instruments Corp). The concentric phase was verified by the vertical displacement calculated from the 4 linear position transducers. As a summary of data calculation, force was calculated as ground reaction force, whereas velocity was measured as changes in vertical displacement of the barbell with respect to time. Then, power was calculated as the product of force and velocity. Mean (MF, MV, and MP) and peak (PF, PV, and PP) variables represent averaged values across all time points and maximum instantaneous values during the concentric phase, respectively. The IMP was measured as an area under the curve of force and time during the concentric phase, whereas RFD was determined at 250 ms (PF at 250 ms divided by 250 ms) from the initiation of the concentric phase (38,39).

Reliability. To evaluate the test-retest reliability of dependent variables across 3 exercise conditions, 2 typical metrics were calculated: intraclass correlation coefficient (ICC) and coefficient of variation (CV). A range of ICC across MF, MP, IMP, PF, PP, and RFD was 0.81–1.00 where RFD (AEL + RR 5) and MF (TS) indicated the lowest and greatest ICC, respectively. The ICC range across MV and PV was relatively lower (0.58–0.84) where MV (AEL + RR 2) and PV (AEL + RR 5) indicated the lowest and greatest ICC, respectively. As for CV, the range across MF, MV, MP, IMP, PF, PV, and PP was 0.34–8.82% in which the lowest and highest CVs were calculated from MF (TS) and PP (AEL + RR 5), respectively. The range of RFD was relatively higher (12.51–30.80%) in which the lowest and highest CVs were calculated from TS and AEL + RR 5, respectively.

Statistical Analyses

An a priori power analysis was conducted using G*Power 3.1.9.7. The result indicated that a total sample size of $n = 3$ is required to obtain an actual power of 0.98 with an assumption that 3 subjects serve as their own control for 3 exercise conditions. Input parameters for the analysis were as follows: $f = 0.25$ (standardized medium effect), $\alpha = 0.05$, power ($1 - \beta$) = 0.85, number of groups = 1, number of measurements = 90, correlation = 0.80, and nonsphericity correction = 1.

All statistical analyses were performed using R (ver. 4.2.1; The R Foundation for Statistical Computing, Vienna, Austria). All data were analyzed using a 3-way repeated-measures analysis of variance (3 conditions by 3 sets by 10 repetitions). Resampling techniques were applied for statistical inference because of main concerns for heteroscedasticity and non-normal distribution of models' residuals. For an omnibus test, permutation was applied to derive p values. When post hoc tests were required, bootstrapping was used to calculate a parameter estimate (mean difference and standardized effect size) and subsequently derive a 95% confidence interval (CI) for each post hoc contrast (pairwise or simple contrasts). The derived 95% CIs were then used for the null hypothesis testing. The following R packages were used: permuco, emmeans, car, simple boot, and boot. Data are reported as mean $\pm$ SD, and an alpha level was $p < 0.05$. Hedge's g effect size magnitudes were interpreted as follows based on Hopkin's guideline (15): 0.0–0.2 (trivial), 0.2–0.6 (small), 0.6–1.2 (moderate), 1.2–2.0 (large), 2.0–4.0 (very large), and >4.0 (nearly perfect).

Results

There was a main effect of condition for MF ($p = 0.007$), PV ($p = 0.049$), and PP ($p = 0.020$) but not for MV ($p = 0.501$), MP ($p = 0.457$), IMP ($p = 0.429$), PF ($p = 0.732$), and RFD ($p = 0.604$). Descriptive statistics, post hoc contrasts, and Hedge's g (95% CI) are shown in Table 3. There was no condition $\times$ set $\times$ repetition interaction or condition $\times$ set interaction for any dependent variables analyzed. However, it was reported that there was a condition $\times$ repetition interaction for MV ($p < 0.001$), MP ($p < 0.001$), IMP ($p < 0.001$), PF ($p < 0.001$), PV ($p = 0.004$), PP ($p = 0.029$), and RFD ($p < 0.001$) but not for MF ($p = 0.704$).

Table 3
Descriptive statistics and effect sizes for each dependent variable.*†‡

	AEL + RR 5	AEL + RR 2	TS
MF (N)	1,706 $\pm$ 224§§	1,697 $\pm$ 209¶	1,685 $\pm$ 211
MV (m·s ⁻¹)	0.68 $\pm$ 0.09	0.66 $\pm$ 0.09	0.67 $\pm$ 0.07
MP (W)	1,157 $\pm$ 231	1,118 $\pm$ 231	1,130 $\pm$ 191
IMP (N·s)	1,688 $\pm$ 348	1,694 $\pm$ 328	1,634 $\pm$ 301
PF (N)	2,671 $\pm$ 474	2,650 $\pm$ 513	2,687 $\pm$ 514
PV (m·s ⁻¹)	1.18 $\pm$ 0.17§§	1.11 $\pm$ 0.19¶	1.10 $\pm$ 0.14
PP (W)	2,304 $\pm$ 499§§	2,148 $\pm$ 512¶	2,079 $\pm$ 388
RFD (N·s ⁻¹)	3,760 $\pm$ 1,610	3,734 $\pm$ 1,653	3,966 $\pm$ 1,668
	AEL + RR 5 AEL + RR 2	AEL + RR 5 TS	AEL + RR 2 TS
MF	0.35 (−0.16 to 0.87)	0.76 (0.66 to 1.95)	0.71 (0.33 to 1.41)
MV	0.25 (−0.36 to 0.69)	0.14 (−0.41 to 0.80)	−0.20 (−0.80 to 0.33)
MP	0.26 (−0.28 to 0.73)	0.25 (−0.25 to 1.01)	−0.09 (−0.70 to 0.40)
IMP	−0.05 (−0.48 to 0.77)	0.32 (−0.21 to 0.96)	0.28 (−0.17 to 0.95)
PF	0.11 (−0.48 to 0.70)	−0.06 (−1.22 to 0.41)	−0.31 (−0.99 to 0.21)
PV	0.38 (−0.04 to 0.79)	0.70 (0.62 to 1.59)	0.16 (−0.34 to 1.06)
PP	0.40 (0.09 to 0.85)	0.80 (0.71 to 1.94)	0.30 (−0.21 to 1.07)
RFD	0.04 (−0.60 to 0.56)	−0.14 (−1.78 to 0.32)	−0.38 (−1.13 to 0.12)

*AEL + RR = combined accentuated eccentric loading and rest redistribution; IMP = impulse; MF = mean force; MP = mean power; MV = mean velocity; PF = peak force; PP = peak power; PV = peak velocity; RFD = rate of force development; TS = traditional sets.

†Data are presented as mean $\pm$ SD collapsed across all sets and repetitions.

‡Effect sizes are presented as between condition Hedge's g (95% confidence interval).

§Significantly different ($p < 0.05$) between AEL + RR 5 and AEL + RR 2.

||Significantly different ($p < 0.05$) between AEL + RR 5 and TS.

¶Significantly different ($p < 0.05$) between AEL + RR 2 and TS.

Figures 2 and 3 present descriptive statistics and post hoc contrasts.

Discussion

The purpose of this study was to investigate acute mechanical responses between 2 AEL + RR conditions and TS during 3 sets of 10 repetitions of BS exercise in resistance-trained men. The authors considered BS exercise, WR, and RR and determined MF, MV, MP, IMP, PF, PV, PP, and RFD to provide coaches with practical and informed programming tactics to optimize strength and power adaptations. In this section, the authors focus on discussing the effect of eccentric/concentric loading and the number of AEL repetitions on potentiation and maintenance of MF, MV, MP, IMP, PF, PV, PP, and RFD across 10 repetitions. This study reveals that AEL + RR 5 and AEL + RR 2 can be practically applied during high-volume BS exercise and informs coaches that AEL + RR 5 may increase especially peak aspects of velocity and power outcomes. This study partially supports our hypotheses and previous literature that used AEL (22,38,39) and RR (1,16,18,23,25–28,34,35).

The AEL + RR 5 and AEL + RR 2 do not potentiate MF given overall $\approx 1\%$ differences in descriptive statistics and a lack of condition by repetition interaction, despite 95% CIs favoring the 2 AEL + RR conditions over TS across 8–9 repetitions. Merrigan et al. (22) used AEL on the initial repetition of a set during 3 sets of 5 repetitions of BS exercise using 120/65% BS 1RM. The authors (22) observed potentiated MF, which was a unique aspect because MF can be increased without using greater “concentric” loading.

The importance of using heavier load has been supported by previous studies where RR maintained MF during 4 sets of 10 repetitions of BS exercise, but did not increase MF (23,25) unless “concentric” loading was increased (4,27). The results of our study indicate that 110/60% BS 1RM and the number of AEL do not cause potentiation but maintenance of MF across 10 repetitions, which was also observed during RR and even TS (4,23,25,27). Collectively, it might be wise that to augment MF, coaches may need to consider prescribing AEL + RR with specifically 120/65% BS 1RM (22) or increasing “concentric” loading (4,27) when AEL is not a feasible choice.

Although previous studies have indicated that MF favors AEL (22), not RR when load was not increased (23,25), MV and MP have been reported to favor AEL (22) and RR (1,16,23,25–27). Nevertheless, our data indicate that 2 AEL + RR conditions do not affect MV and MP, causing no differences at 8–10 repetitions compared with TS. Merrigan et al. (22) showed that 120/65% BS 1RM potentiated MV and MP, with greater values until the third to fourth repetition in a set of 5 compared with TS. Wagle et al. (38,39) that used 105/80% loading did not find the potentiation of MV, despite the same number of AEL repetition ($n = 1$), underlining the importance of eccentric/concentric loading magnitude. Researchers have reported that RR maintained MV and MP, with greater values at later 4–5 cluster repetitions compared with TS (1,16,23,25–27). The maintenance effect of MV and MP was shown to be less when 6 cluster repetitions were used (18), emphasizing the importance of the number of cluster repetitions. Despite the consideration of those 2 aspects to fully benefit from the potentiation (22,38,39) and maintenance (1,16,18,23,25–27), our

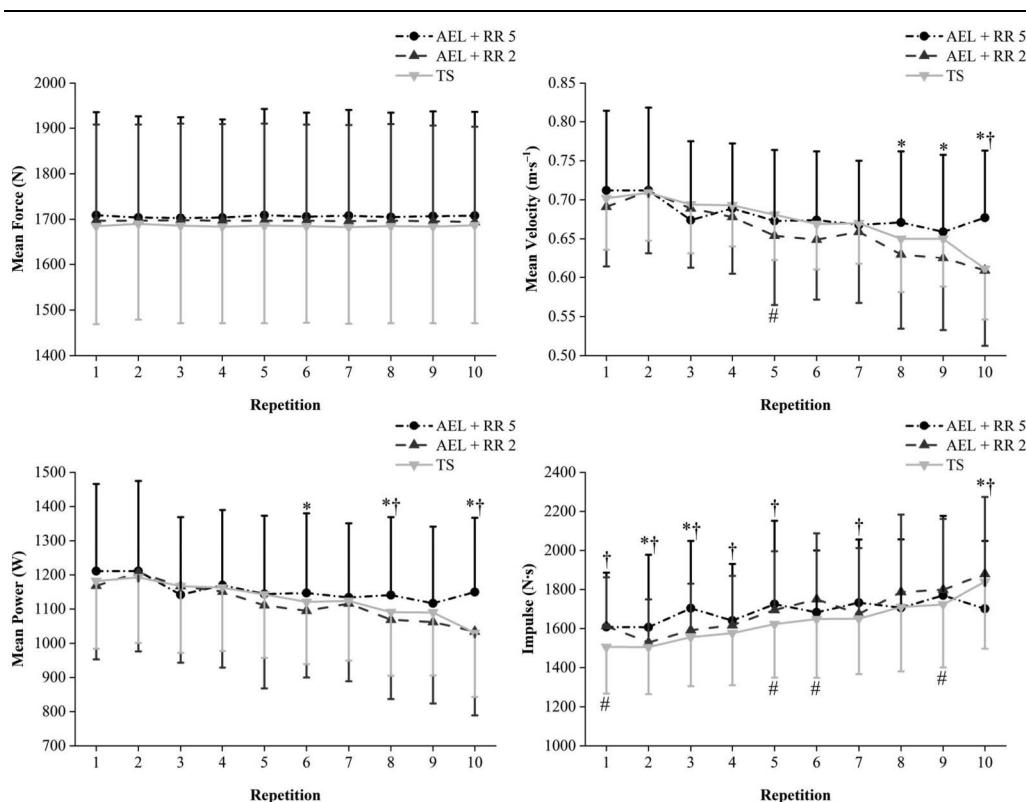


Figure 2. Significant differences in mean values and impulse across conditions. Data are presented as mean $\pm$ SD. *Significantly different ($p < 0.05$) between AEL + RR 5 and AEL + RR 2. †Significantly different ($p < 0.05$) between AEL + RR 5 and TS. #Significantly different ($p < 0.05$) between AEL + RR 2 and TS. AEL + RR = combined accentuated eccentric loading and rest redistribution; TS = traditional sets.

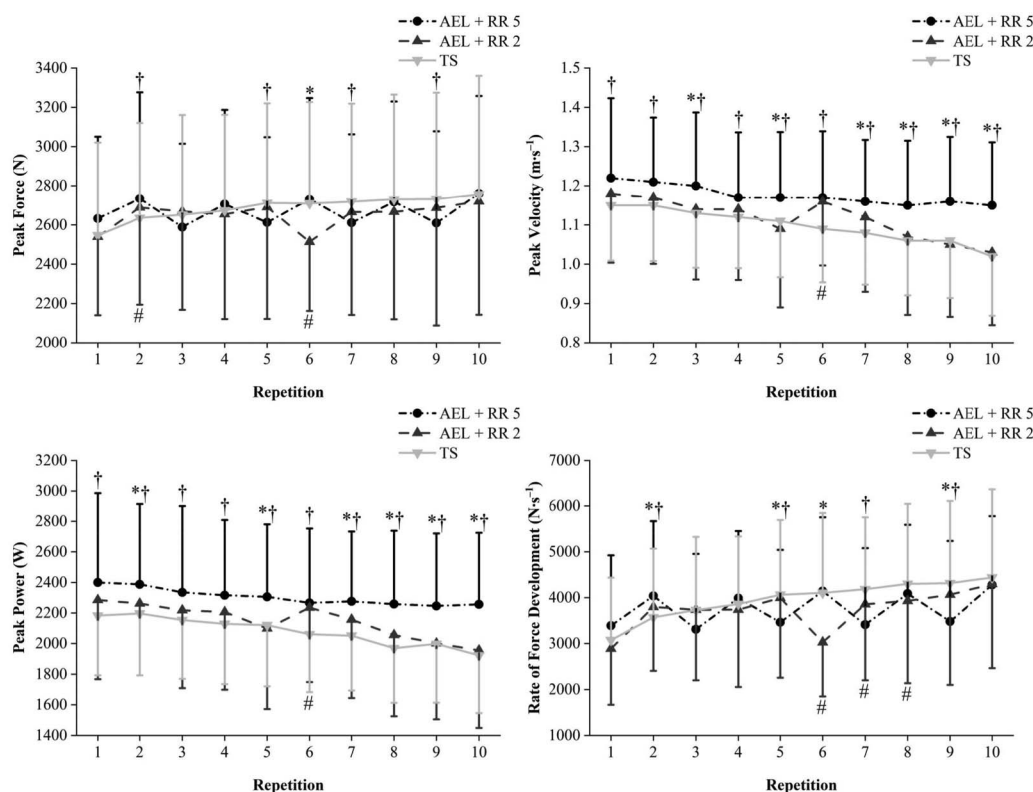


Figure 3. Significant differences in peak values and rate of force development across conditions. Data are presented as mean $\pm$ SD. *Significantly different ($p < 0.05$) between AEL + RR 5 and AEL + RR 2. †Significantly different ($p < 0.05$) between AEL + RR 5 and TS. #Significantly different ($p < 0.05$) between AEL + RR 2 and TS. AEL + RR = combined accentuated eccentric loading and rest redistribution; TS = traditional sets.

experimental conditions have no effect on MV and MP, suggesting 3 possible explanations. First, 110/60% BS 1RM might not have met a threshold of eccentric loading or a gap between eccentric and concentric loading to potentiate MV and MP compared with 120/65% loading (22). Second, because cadence of eccentric phase was not prescribed, some subjects' slow descent pace strategy during eccentric phase might have dissipated potentiation compared with a faster descent strategy (20–22). Third, 15-second intraset rest might not have been long to maintain MV and MP compared with 30–60 seconds (1,16,23,25–27). Future research should address the 3 factors to shed light on potentiation and maintenance of MV and MP.

To the best of authors' knowledge, no researchers have previously examined IMP responses in BS exercise using AEL or RR. As IMP is the product of force and time, it has been considered a metric with an excellent relevance to strength and power adaptations (8,36,37). In our study, the responses of IMP favor AEL + RR 5 and AEL + RR 2 over TS at 4–6 repetitions in discussing a condition $\times$ repetition interaction. However, in discussing a main effect of condition (overall effects), statistical differences were not inferred with small effect sizes and $\approx$ 3–4% differences in descriptive statistics between 2 AEL + RR conditions and TS, all of which indicate negligible effects. Despite the trivial differences in our study and the lack of data from previous works, changes in IMP across 10 repetitions observed provide an important implication for IMP during high-volume exercise. The IMP seems to increase with repetitions because of decrease in MV but does not seem to be exactly opposite to MV because MF is different, albeit trivial, between conditions. It is not surprising because of the

equation of IMP (the product of force and time). However, given that the “time” is inversely related to MV, it might not be advisable to pursue potentiation and maintenance of MV when coaches aim to increase IMP as a main training goal. Instead, seeking a programming tactic that increases MF might be desirable [e.g., increasing “concentric” loading (4,27)] because MV inherently decreases across 10 repetitions.

As discussed throughout this section, potentiation of MF, MV, and MP seems to be highly sensitive to eccentric/concentric loading magnitude (22,38,39). An examination of PF, PV, and PP may also confer an insight into the effect of eccentric/concentric loading magnitude on potentiation of peak variables. Interestingly, our study indicates that 110/60% BS 1RM and the number of AEL repetitions compromise PF and RFD, leading to lower values at later repetitions for which AEL is used compared with TS. The lower values during AEL + RR 5 and AEL + RR 2 might be explained in part by the observation that TS (statistically) increase PF and RFD from the second to tenth repetitions compared with the first repetition. Our data do not agree with those of Merrigan et al. (22) who reported that 120/65% BS 1RM resulted in greater PF until the third repetition in a set compared with TS, but agree with those of Wagle et al. (38,39) who reported that 105/80% loading did not produce greater RFD. The results of our study disagree with previous literature where there were no differences in PF between RR and TS that used the same or different concentric loading (4,23). The inconsistency in PF between this study and previous works (4,22,23) may be explained by that 110/60% BS 1RM might have decreased peak acceleration, given that concentric loading remained constant (i.e., force is mass times

acceleration). However, it is important to note that 110/60% BS 1RM does not impact MF but PF, indicating that mean and peak values respond to eccentric overload in a unique manner.

Supporting this contention, our data suggest that 110/60% BS 1RM and the number of AEL repetitions impact potentiation and maintenance of PV and PP. This is explained by the observation that AEL + RR 5 induces greater PV and PP across 10 repetitions, whereas AEL + RR 2 causes greater values only at the sixth repetition compared with TS. The elevated PV and PP across all 10 repetitions during AEL + RR 5 have not been demonstrated in any previous literature that has prescribed AEL or RR. Investigators reported that 120/65% and 105/80% BS 1RM failed to potentiate PV (22) and PP (22,38,39) and also that RR maintained PV and PP, with greater values only at later 1–3 of 5 cluster repetitions when compared with TS (23). The AEL + RR 5 potentiates PV and PP at early repetitions, despite the shorter inter-set rest and maintains PV and PP to a greater level from the first to tenth repetitions compared with TS (94 vs. 88–89%), resulting in overall ≈ 7 –11% differences in descriptive statistics. The greatest PV and PP observed during AEL + RR 5 support our hypothesis because of more frequent potentiation and maintenance across 10 repetitions. Although the increase in PV and PP across all 10 repetitions seems to be highly unique and promising, coaches might need to be aware that such acute responses may not chronically improve lower-body power development (29). Given that PV and PP occur much later than a sticking region (i.e., at the end of the concentric “acceleration” phase), their practical relevance to power development might not be as great as MV and MP that represent the entire concentric phase (29). Nevertheless, our data still provide a unique insight into the effect of 110/60% BS 1RM and the number of AEL repetitions on peak kinetic and kinematic responses where PV and PP can be elevated using AEL + RR 5 even with decreased PF.

To conclude, our results demonstrate that mean and peak kinetics and kinematics are highly sensitive to eccentric overload magnitude. The effect of 110/60% BS 1RM and the number of AEL repetitions on MF and IMP is negligible, given overall ≈ 1 and 3–4% differences, respectively, when compared with TS. The 2 AEL + RR conditions do not potentiate and maintain MV and MP and even compromise PF and RFD at later repetitions for which AEL is used. However, AEL + RR 5 induces the greatest PV and PP among all 3 conditions with ≈ 7 –11% greater values when compared with TS.

Practical Applications

Our results suggest that applying eccentric overload to every other repetition could increase especially peak aspects of velocity and power outcomes across 10 repetitions by ≈ 7 –11%. Coaches may prescribe this advanced method in a strength-endurance phase of block periodization where a traditional strategy compromises those specific mechanical outcomes during high-volume exercise.

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